

What Is Hypothesis Testing in Statistics?

Hypothesis Testing is a type of statistical analysis in which you put your assumptions about a population parameter to the test. It is used to estimate the relationship between 2 statistical variables.

Null Hypothesis and Alternative Hypothesis

The Null Hypothesis is the assumption that the event will not occur. A null hypothesis has no bearing on the study's outcome unless it is rejected.

The Alternate Hypothesis is the assumption that is to be proved. The acceptance of the alternative hypothesis follows the rejection of the null hypothesis. H_1 is the symbol for it.

Hypothesis Testing Formula

$$Z = (\bar{x} - \mu_0) / (\sigma / \sqrt{n})$$

- Here, $\bar{x}$ is the sample mean,
- μ_0 is the population mean,
- σ is the standard deviation,
- n is the sample size.

How Hypothesis Testing Works?

An analyst performs hypothesis testing on a statistical sample. Measurements and analyses are conducted on a random sample of the population to test two hypotheses: the null and alternative hypotheses.

Steps in Hypothesis Testing

Formulate Hypotheses

- Null Hypothesis (H_0): This hypothesis states that there is no effect or difference, and it is the hypothesis you attempt to reject with your test.
- Alternative Hypothesis (H_1): This hypothesis is what you might believe to be true or hope to prove true. It is usually considered the opposite of the null hypothesis.

Choose the Significance Level (α)

The significance level, often denoted by **alpha (α)**, is the probability of rejecting the null hypothesis when it is true. Common choices for α are 0.05 (5%), 0.01 (1%), and 0.10 (10%).

Select the Appropriate Test

Choose a statistical test based on the type of data and the hypothesis. Common tests include **t-tests, chi-square tests, ANOVA, and regression analysis**. The selection depends on data type, distribution, sample size, and whether the hypothesis is one-tailed or two-tailed.

Collect Data

Gather the data that will be analyzed in the test. To infer conclusions accurately, this data should be representative of the population.

Calculate the Test Statistic

Based on the collected data and the chosen test, calculate a test statistic that reflects how much the observed data deviates from the null hypothesis.

Determine the p-value

The p-value is the probability of observing test results at least as extreme as the results observed, assuming the null hypothesis is correct. It helps determine the strength of the evidence against the null hypothesis.

Make a Decision

Compare the p-value to the chosen significance level:

- If the $p\text{-value} \leq \alpha$: Reject the null hypothesis, suggesting sufficient evidence in the data supports the alternative hypothesis.
- If the $p\text{-value} > \alpha$: Do not reject the null hypothesis, suggesting insufficient evidence to support the alternative hypothesis.

Types of Hypothesis Testing

1. Z Test

To determine whether a discovery or relationship is statistically significant, hypothesis testing uses a z-test. It usually checks to see if two means are the same (the null hypothesis). Only when the population standard deviation is known and the sample size is 30 data points or more, can a z-test be applied.

2. T Test

The “t-test” or “Student’s t-test” developed by William Sealy Gosset, (who wrote under the pseudonym “Student”) compares the means of two groups to see if they are significantly different from each other.

There are different types of t-tests, such as:

- Independent t-test: Compares the means of two independent groups.
- Paired t-test: Compares means from the same group at different times.
- One-sample t-test: Compares the mean of a single group against a known mean.

It is frequently used in hypothesis testing.

3. Chi-Square

Chi-square test is used for hypothesis testing to determine if the expected and observed results are well-fitted. The Chi-square test analyzes the differences between categorical variables from a random sample. The test's fundamental premise is that the observed values in the data should be compared to the predicted values that would be present if the null hypothesis were true.

4. ANOVA

ANOVA, or Analysis of Variance, is a statistical method used to compare **two or more groups**. It's particularly useful when there are significant differences between multiple groups. For instance, in business, a company might use ANOVA to analyze whether three different stores are performing differently in terms of sales. It's widely used in fields like medical research and social sciences, where comparing group differences can provide valuable insights.

Type 1 and Type 2 Error

A hypothesis test can result in two types of errors.

Type 1 Error: A Type-I error occurs when sample results reject the null hypothesis despite being true.

Type 2 Error: A Type-II error occurs when the null hypothesis is not rejected when it is false, unlike a Type-I error.

Example Problems on Hypothesis Testing

Here are the practice problems on hypothesis testing that will help you understand how to apply these concepts in real-world scenarios:

Example:

The mean marks scored by students in an examination was 67.39 with standard deviation 1.30. Then students were taught using a new teaching methodology. After being taught in new teaching methodology students appeared in a similar examination. The mean marks scored by a sample of 60 students was found to be 67.47. Analyse the usefulness of new teaching method.

Solution:

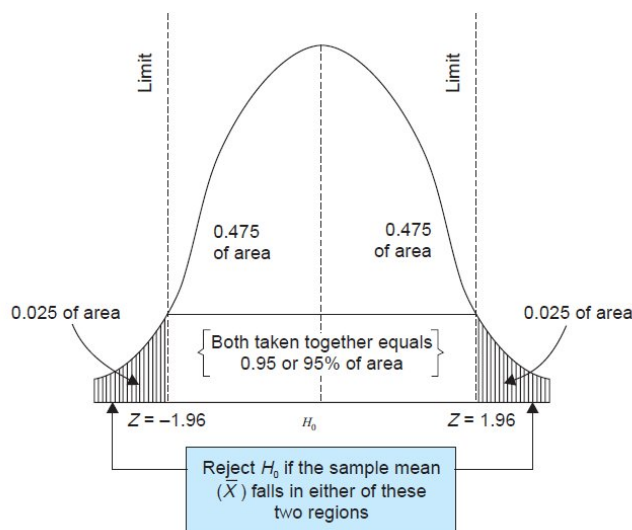
H_0 : There is no effect of new teaching method

H_a : The new teaching method is effective in improving the marks of students.

Given information: $\mu = 67.39$ $\sigma_p = 1.30$, $\bar{X} = 67.57$, $n = 60$.

$$z = \frac{\bar{X} - \mu}{\sigma_p / \sqrt{n}} = (67.57 - 67.39) / (1.30 / \sqrt{60}) = 1.07$$

P value corresponding to $z=1.07$ is 0.142

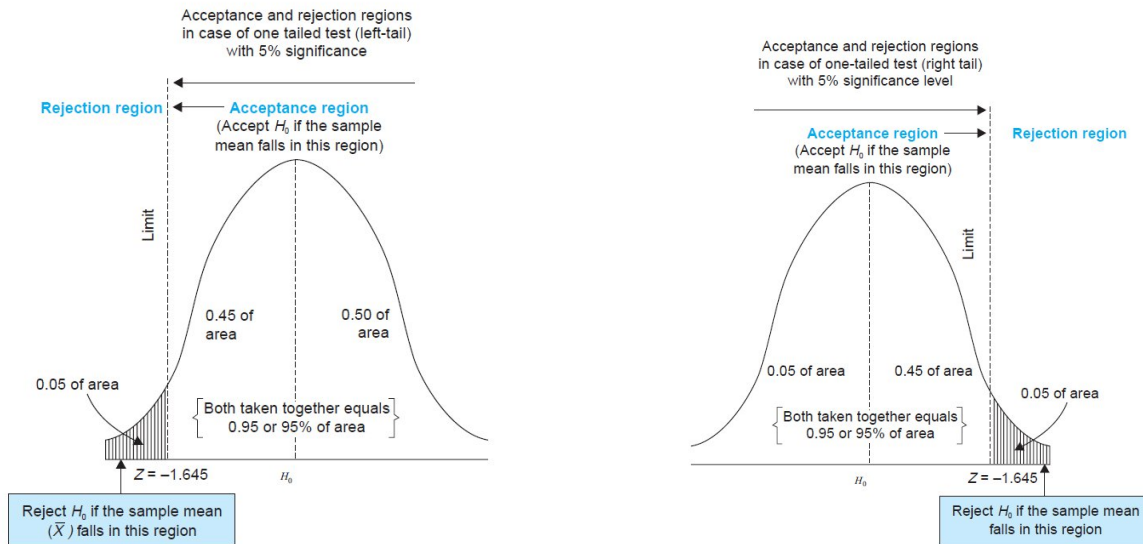


Since

$-1.96 < 1.07 < 1.96$ i.e. p value $0.142 > 0.05$ the null hypothesis is not rejected.

H_0 : The new teaching method is not effective in improving the marks of students.

H_a : The new teaching method is effective in improving the marks of students.



Since $1.07 < 1.645$ i.e. p value $0.142 > 0.05$ the null hypothesis is not rejected.

By not rejecting the null hypothesis, we might be committing type-II error. It is possible that the new teaching method is actually effective in improving marks of students.

Prob. Corresponding to $z = 1.07$ is 0.8577

Prob (improvement) – 14.23%

Type-II error $\beta = 0.1423$

Example:

The mean marks scored by students in an examination was 67.39 with standard deviation 1.30. Then students were taught using a new teaching methodology. After being taught in new teaching methodology students appeared in a similar examination. The mean marks scored by a sample of 50 students was found to be 67.90. Analyse the usefulness of new teaching method.

$$z = \frac{\bar{X} - \mu}{\sigma_p / \sqrt{n}} = (67.90 - 67.39) / (1.30 / \sqrt{50}) = 2.77$$

p value corresponding to $z = 2.77$ is 0.0028

Since $2.77 > 1.645$ ie p value $0.0028 < 0.05$ the null hypothesis is rejected.

H_0 : The new teaching method is not effective in improving the marks of students.

By rejecting the null hypothesis, we might be committing type-I error. It is possible that the new teaching method is actually not effective in improving marks of students.

Type-I error $\alpha = 0.0028$

Power of test is $(1 - \beta) = 1 - 0.0028 = 99.77\%$